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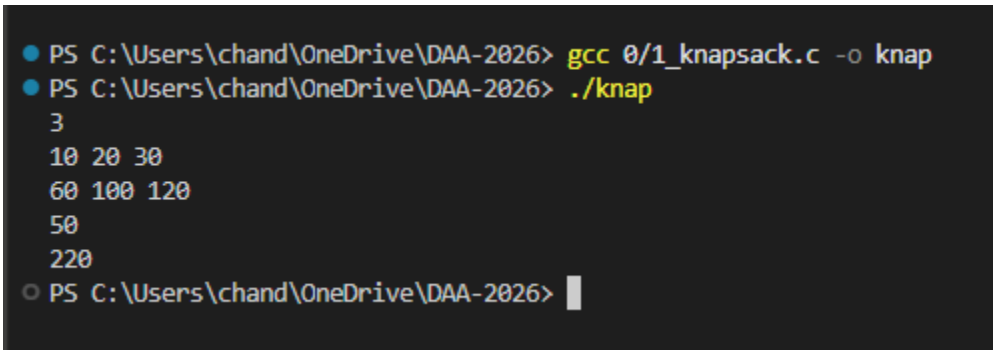
AP25110010976

1. Dynamic Programming: Implement 0/1 Knapsack problem.

Program:

```
#include<stdio.h>
int max(int a,int b)
{
return (a>b)?a:b;
}
int main()
{
int n,W,i,w;
scanf("%d",&n);
int wt[n],val[n];
for(i=0;i<n;i++)
scanf("%d",&wt[i]);
for(i=0;i<n;i++)
scanf("%d",&val[i]);
scanf("%d",&W);
int dp[n+1][W+1];
for(i=0;i<=n;i++)
{
for(w=0;w<=W;w++)
{
if(i==0||w==0)
dp[i][w]=0;
else if(wt[i-1]<=w)
dp[i][w]=max(val[i-1]+dp[i-1][w-wt[i-1]],dp[i-1][w]);
else
dp[i][w]=dp[i-1][w];
}
}
printf("%d\n",dp[n][W]);
return 0;
}
```

Terminal Process: